



Installation Instruction

Installation Instruction Number: 4908441

Released Date: 12-may-2015

Aftertreatment Outlet NOx Sensor Internal Water Shield

Aftertreatment Outlet NOx Sensor Internal Water Shield

The purpose of this document is to guide the user through the installation steps for the aftertreatment outlet NOx sensor internal water shield.

Product Affected

- ISB6.7 CM2350 B101
- ISB6.7 CM2350 B103
- ISL9 CM2350 L101
- ISL9 CM2350 L111
- ISX12 CM2350 X102
- ISX15 CM2350 X101

Service Parts

The parts required for installation are shown in Table 1 and 2 below.

Table 1, Service Parts for ISX12, ISX15

Description	Part Number	Quantity
Drip Shield	4352880	1
Screw, Hex Flange Head Cap	4354269	2
Template	4353074	1
High Temp Sealant	5362108	1

Table 2, Service Parts for ISB6.7, ISL9

Description	Part Number	Quantity
Drip Shield	4353926	1
Screw, Hex Flange Head Cap	4354269	2
Template	4353074	1
High Temp Sealant	5362108	1

Cummins® High Temp Sealant, Part Number 5362108, is the **only** sealant approved to be used in this procedure.

A video of the following installation procedure can be found in the electronic version of this Installation Instruction located on QuickServe® Online. Use the following link to view the video.

Note : <http://tsb.cumminsvirtualcollege.com/TSB150057.aspx>
(<http://tsb.cumminsvirtualcollege.com/TSB150057.aspx>)

Required Tools

8 mm [5/16 in] Drill Bit

End Wrench

Center Punch

Cleaning Solvent

Electric Drill

Flathead Screwdriver

Masking Tape

Metal File

Needle Nose Pliers

Rubber Gloves

Ruler Measuring 6 Inches or Greater

Safety Glasses

Sand Paper 40 to 60 Grit

Shop Towels

Stir Stick

Aftertreatment Selective Catalytic Reduction (SCR) Catalyst and Aftertreatment Outlet NOx Sensor Removal

⚠ WARNING ⚠

Wait for the aftertreatment system to cool down before working on the hot exhaust or exhaust components. Failure to do so could result in personal injury.

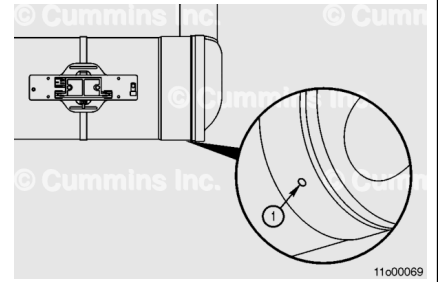
- Remove aftertreatment SCR assembly from chassis. See Aftertreatment Selective Catalytic Reduction (SCR) Catalyst, Procedure 011-036 in Section 11, in the corresponding Service Manual.
- Remove aftertreatment outlet NOx sensor from aftertreatment SCR. See Aftertreatment Outlet NOx Sensor, Procedure 019-451 in Section 19, in the corresponding Service Manual.



Aftertreatment SCR Catalyst Assembly Drain Hole Inspection

⚠ CAUTION ⚠

Do not use a metallic object to clean the aftertreatment SCR catalyst. This will scratch the surface of the aftertreatment SCR catalyst which may cause excessive diesel exhaust fluid buildup.

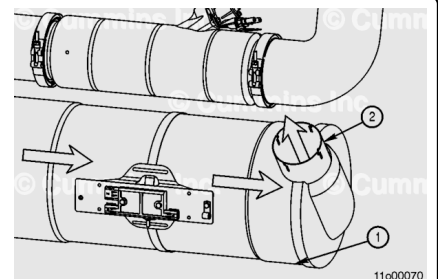


⚠ CAUTION ⚠

Do not use any chemical solvents or cleaners to clean internal metal surface. Possible damage to aftertreatment SCR catalyst may occur.

- Every aftertreatment SCR catalyst unit has a drain hole (1) which allows water accumulated in the unit to drain out of the system. A plugged drain hole can cause an aftertreatment outlet NOx sensor to malfunction. The drain hole **must** be clear of debris, dirt, or any other material.
- If debris, dirt, or any other material is present carefully clean the affected area with a non-metallic tool.

- The drain hole (1) is **always** oriented in the lowest section of the aftertreatment SCR catalyst outlet (2), pointed toward ground.



Drilling Holes In Aftertreatment SCR Catalyst Outlet Pipe

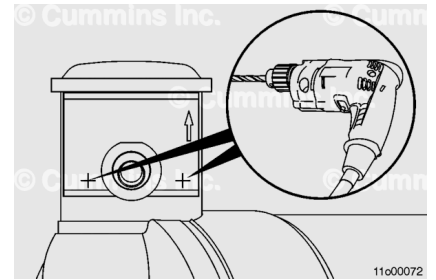
- Cut template for either a Marmon joint or Torca joint. See template, Part Number 4353074, for more information.
- Attach appropriate template to aftertreatment SCR catalyst outlet pipe.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear goggles and protective clothing.

- Use center punch to mark holes at designated location on template.
- Drill holes using bit sizes specified on template.
- Remove template.



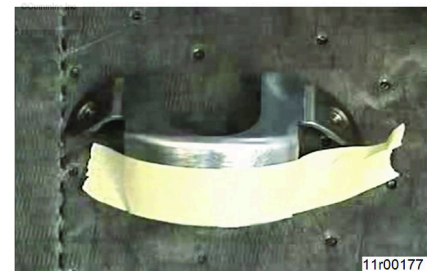
Cleaning and Preparing Surfaces

- Remove any burrs from holes using a metal file.
- Clean metal shavings out of aftertreatment SCR catalyst. These shavings could potentially clog drain hole.
- Mask off sensing wheel cup screen area using masking tape.
- Roughen the smooth surfaces of the SCR catalyst pipe on either side of the sensing wheel and directly above the sensing wheel using 40 to 60 grit sand paper.
- Roughen the top (the side facing up while looking down into the SCR catalyst outlet) of the outlet NOx sensor internal water shield's surface using 40 to 60 grit sand paper.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

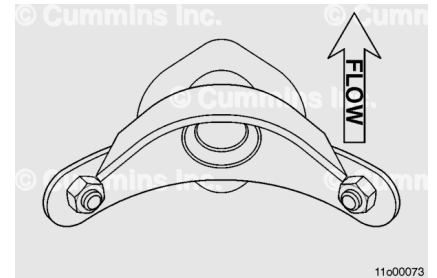
Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.

- Use cleaning solvent to clean surfaces thoroughly from oil, grease, dirt, corrosion, or other contaminants.

- Attach masking tape to aftertreatment outlet NOx sensor internal water shield, making sure to leave one inch of exposed metal as pictured.

Aftertreatment Outlet NOx Sensor Internal Water Shield Installation

- Orient aftertreatment outlet NOx sensor internal water shield towards aftertreatment SCR catalyst outlet. See illustration on right with exhaust flow direction arrow.
- Install two capscrews and washers from outside and hand-tighten.
- Tighten capscrews.



Torque Value: 7.3 n•m [5.4 ft-lb]

- Attach masking tape to aftertreatment SCR catalyst outlet pipe. Apply masking tape 1 inch high and 4 inches wide about 1 inch above the outlet NOx sensor internal water shield. The purpose of this masking tape is to remove excess high temp sealant.



Cummins® High Temp Sealant Application

⚠ CAUTION ⚠

Do not allow high temp sealant to fall into the aftertreatment SCR catalyst during application.

- The high temp sealant **must** be mixed thoroughly for at least one minute.
- All surfaces **must** be free of oil, grease, dirt, corrosion, or other contaminants before using high temp sealant.
- Smooth metal surfaces **must** be roughened using 40 to 60 grit sand paper before using high temp sealant.



- With rubber gloves, use a finger or an application tool to apply the high temp sealant evenly to seal the gap between aftertreatment SCR catalyst outlet pipe and aftertreatment outlet NOx sensor internal water shield.
- Make sure the high temp sealant is well packed. If the high temp sealant is **not** well packed, the bonding will **not** be

good and air pockets can result.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear goggles and protective clothing.

⚠ CAUTION ⚠

Do not allow high temp sealant to fall into the aftertreatment SCR catalyst during masking tape removal.



- The masking tape **must** be removed before the high temp sealant hardens. At room temperature the high temp sealant will harden in around 4 hours.
- Remove excess high temp sealant using a rag or a gloved hand.
- Remove masking tape.

Finishing Steps

- Install aftertreatment outlet NOx sensor . See Aftertreatment Outlet NOx Sensor, Procedure 019-451 in Section 19, in the corresponding Service Manual.
- Install aftertreatment SCR catalyst assembly. See Aftertreatment Selective Catalytic Reduction (SCR) Catalyst, Procedure 011-036 in Section 11, in the corresponding Service Manual.
- To cure the high temp sealant, run a stationary regeneration after installing the aftertreatment SCR catalyst assembly. See Aftertreatment Diesel Particulate Filter (DPF) Regeneration Test, Procedure 014-016 in Section 14, in the corresponding Service Manual.

©Cummins Inc.

Document History

Date	Details
2015-5-12	Module Created

Last Modified: 05-Jun-2015
